Quantitative Research

MOCK EXAM 1

Practice exam for exam preparation – not the actual module examination

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First: This exam consists of 3 tasks with 10 subtasks each. Please check that your copy is complete.

Time allowed: 120 minutes.

Not allowed: The staple of the exam must not be removed.

Not allowed: Copying other people's solutions and using aids that are not permitted.

Allowed: The only permitted aids are one of the calculators named in the lecture (Casio fx-85-MS, fx-85-MS-2, fx-82-MS or fx-82-MS-2) and a handwritten formula sheet of at most one A4 sheet.

Points: Each subtask is worth **2 points**; each task gives 20 points, the whole exam **60 points**.

Independence: All subtasks can be solved independently of each other. Every subtask states all the numbers needed to solve it. If you cannot solve a subtask, simply continue with the others – you lose nothing else.

Note 1: Each subtask asks for exactly *one* result. Write your working and the result on the pages with a correction margin after each task, and mark the result clearly.

Note 2: A result without comprehensible working earns only half of the points.

Note 3: Round final results to four decimal places, or to two where this is sensible (e.g. amounts in euros).

Python: Some subtasks ask for a Python command or for the value that a given line of Python code returns. Answer them by hand – no computer is needed.

Tables: The standard normal table (Table 1) and the t-table (Table 2) are at the end of the exam.



Good luck!

Task 1 (20 points – 10 subtasks worth 2 points each)
An online shop for sports equipment plans with a mean order value of $\mu_0 = 80$ EUR. From many years of experience, the standard deviation of the order values is known: $\sigma = 12$ EUR. After a redesign of the website, a random sample of $n = 36$ orders gives a mean order value of $\bar{x} = 84.30$ EUR. The significance level is $\alpha = 0.05$. Because σ is known, the test and the confidence interval are based on the standard normal distribution (Table 1).

All subtasks can be solved independently of each other.

a) The management wants to find out whether the mean order value after the redesign *differs* from the planning figure of 80 EUR (in either direction). State the hypothesis pair. (2 points)

b) Compute the standard error $\sigma/\sqrt{n}$ of the sample mean for $\sigma = 12$ EUR and $n = 36$. (2 points)

c) Compute the test statistic Z_n for $\bar{x} = 84.30$ EUR, $\mu_0 = 80$ EUR, $\sigma = 12$ EUR and $n = 36$. (2 points)

d) In Python, the critical value of the two-sided test at the significance level $\alpha = 0.05$ (σ known) is computed with the line below. Which number must be passed as `q`?

```
stats.norm.ppf(q)
```

(2 points)

e) The test statistic is $Z_n = 2.15$ and the critical value is $z_{0.975} = 1.96$. Make the test decision for $H_0\colon \mu = 80$ vs. $H_1\colon \mu \neq 80$ at $\alpha = 0.05$. (2 points)

f) Compute the two-sided p-value for the test statistic $Z_n = 2.15$ (use Table 1). (2 points)

g) Compute the 95 % confidence interval for μ with $\bar{x} = 84.30$ EUR, $\sigma/\sqrt{n} = 2$ EUR and $z_{0.975} = 1.96$. (2 points)

h) For the mean order value μ, the 95 % confidence interval $[80.38;\ 88.22]$ EUR was computed. Interpret this interval in one sentence. (2 points)

i) Describe in one sentence what a Type I error would mean in this situation. (2 points)

j) In practice σ is often unknown. Suppose the 36 order values are stored in a NumPy array `orders` and σ is estimated from the data. Give the `scipy.stats` command that carries out the two-sided one-sample t-test of $H_0\colon \mu = 80$. (2 points)

Task 2 (20 points – 10 subtasks worth 2 points each)
Part A (subtasks a–h). A café records on five summer days the daily maximum temperature x (in °C) and the number y of iced coffees sold:

Day i	1	2	3	4	5
Maximum temperature x_i (°C)	20	22	24	26	28
Iced coffees sold y_i	46	54	62	70	68

Model: $y_i = \beta_0 + \beta_1 x_i + u_i$. Auxiliary values:

$$\bar{x} = 24, \qquad \bar{y} = 60, \qquad S_{xx} = \sum (x_i - \bar{x})^2 = 40,$$
$$S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) = 120, \qquad S_{yy} = \sum (y_i - \bar{y})^2 = 400.$$

Part B (subtasks i–j). For a larger sample of $n = 60$ summer days the café also records the rainfall `rain` (in mm) and estimates a multiple linear regression of `sales` (iced coffees sold) on `temp` (maximum temperature in °C) and `rain`. Excerpt of the `statsmodels` output:

```
Dep. Variable:                  sales   R-squared:                       0.947
Model:                            OLS   Adj. R-squared:                  0.945
Method:                 Least Squares   F-statistic:                     512.4
                  coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept    -12.2152      3.783     -3.229      0.002     -19.790      -4.641
temp           3.0055      0.131     22.863      0.000       2.742       3.269
rain          -0.4313      0.374     -1.154      0.253      -1.180       0.317
```

All subtasks can be solved independently of each other.

a) Compute the slope $\hat{\beta}_1$ from $S_{xy} = 120$ and $S_{xx} = 40$. (2 points)

b) Compute the intercept $\hat{\beta}_0$ from $\hat{\beta}_1 = 3$, $\bar{x} = 24$ and $\bar{y} = 60$. (2 points)

c) Interpret the slope $\hat{\beta}_1 = 3$ in one sentence in the context of the café. (2 points)

d) The fitted regression of `sales` on `temp` is stored in the Python object `reg`; its estimated line is $\hat{y} = -12 + 3x$. Which value does the following line return?

```
reg.predict(pd.DataFrame({"temp": [25]}))
```
(2 points)

e) Compute the residual $\hat{u}_5$ of day 5 with $(x_5;\ y_5) = (28;\ 68)$, using the estimated regression line $\hat{y} = -12 + 3x$. (2 points)

f) For the five days, the total sum of squares is TSS $= 400$ and the residual sum of squares is RSS $= 40$. Compute the coefficient of determination R^2. (2 points)

g) Determine the correlation coefficient r_{xy} from $R^2 = 0.9$ and the slope $\hat{\beta}_1 = 3$. (2 points)

h) The five days are stored in a pandas DataFrame `cafe` with the columns `temp` and `sales`. Give the `statsmodels` command that estimates the simple linear regression of `sales` on `temp`. (2 points)

i) In the multiple regression of Part B, the coefficient of `temp` is $\hat{\beta}_1 = 3.0055$. Interpret this coefficient in one sentence. (2 points)

j) In the multiple regression of Part B, the coefficient of `rain` is -0.4313 with the p-value 0.253 (column `P>|t|`). Decide at $\alpha = 0.05$ whether rainfall has a significant effect on sales when the temperature is held fixed. (2 points)

Task 3 (20 points – 10 subtasks worth 2 points each)
A mobile-phone provider wants to predict whether a customer cancels the contract in the next quarter ($Y = 1$: churn, $Y = 0$: no churn). The explanatory variable x is the number of calls to the service hotline in the last six months. A logistic regression gives the estimated linear predictor (log-odds)

$$\hat{\eta}(x) = \hat{\beta}_0 + \hat{\beta}_1 x = -3 + 0.5\,x \qquad \text{and} \qquad \hat{p}(x) = \frac{e^{\hat{\eta}(x)}}{1 + e^{\hat{\eta}(x)}}.$$

The provider then classifies $n = 200$ test customers with the threshold $t = 0.5$ (predicted churn if $\hat{p}(x) > 0.5$):

	Pred. No	Pred. Yes	Total
Actual No	TN = 140	FP = 20	160
Actual Yes	FN = 10	TP = 30	40
Total	150	50	200

All subtasks can be solved independently of each other.

a) Last year, 20 % of all customers churned, i.e. $p = 0.2$. Compute the odds of churn. (2 points)

b) Compute the predicted churn probability $\hat{p}(x)$ of a customer with $x = 2$ hotline calls, using $\hat{\eta}(x) = -3 + 0.5\,x$ and $e^{-2} = 0.1353$. (2 points)

c) The odds ratio of the hotline calls is $e^{\hat{\beta}_1} = e^{0.5} = 1.6487$. Interpret it in one sentence. (2 points)

d) A customer has $x = 7$ hotline calls. Classify this customer with the threshold $t = 0.5$, using $\hat{\eta}(x) = -3 + 0.5\,x$ (you may use $e^{0.5} = 1.6487$). (2 points)

e) The customer data are stored in a pandas DataFrame `df` with the columns `churn` (coded 0/1) and `calls`. Give the `statsmodels` command that estimates the logistic regression of `churn` on `calls`. (2 points)

f) Compute the accuracy of the classifier from the confusion matrix with TN = 140, FP = 20, FN = 10 and TP = 30. (2 points)

g) The DataFrame `test` holds the actual class `churn` (coded 0/1) of the test customers, `phat` their predicted churn probabilities; at $t = 0.5$ the confusion matrix has TN = 140, FP = 20, FN = 10 and TP = 30. Which value does the following line show in the row `churn` = 1, column `True`?

```
pd.crosstab(test.churn, phat > 0.5, normalize="index")
```

(2 points)

h) Compute the specificity of the classifier from the confusion matrix with $\text{TN} = 140$, $\text{FP} = 20$, $\text{FN} = 10$ and $\text{TP} = 30$. (2 points)

i) Compute the precision of the classifier from the confusion matrix with $\text{TN} = 140$, $\text{FP} = 20$, $\text{FN} = 10$ and $\text{TP} = 30$. (2 points)

j) To catch more churners, the provider lowers the threshold from $t = 0.5$ to $t = 0.3$. Does the sensitivity increase or decrease? (2 points)

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Reading: row $=z$ to one decimal, column $=$ second decimal, e.g. $\Phi(1.96) = 0.9750$. Negative z: $\Phi(-z) = 1 - \Phi(z)$. For $z \geq 3.5$: $\Phi(z) \approx 1$.

Selected quantiles z_p with $\Phi(z_p) = p$					
p	0.90	0.95	0.975	0.99	0.995
z_p	1.282	1.645	1.960	2.326	2.576

Table 1: Cumulative distribution function $\Phi(z) = P(Z \leq z)$ of the standard normal distribution $N(0,1)$

	probability p (quantile $t_{p;\,\mathrm{df}}$)					
df	0.80	0.90	0.95	0.975	0.99	0.995
upper tail	0.20	0.10	0.05	0.025	0.01	0.005
1	1.3764	3.0777	6.3138	12.7062	31.8205	63.6567
2	1.0607	1.8856	2.9200	4.3027	6.9646	9.9248
3	0.9785	1.6377	2.3534	3.1824	4.5407	5.8409
4	0.9410	1.5332	2.1318	2.7764	3.7469	4.6041
5	0.9195	1.4759	2.0150	2.5706	3.3649	4.0321
6	0.9057	1.4398	1.9432	2.4469	3.1427	3.7074
7	0.8960	1.4149	1.8946	2.3646	2.9980	3.4995
8	0.8889	1.3968	1.8595	2.3060	2.8965	3.3554
9	0.8834	1.3830	1.8331	2.2622	2.8214	3.2498
10	0.8791	1.3722	1.8125	2.2281	2.7638	3.1693
11	0.8755	1.3634	1.7959	2.2010	2.7181	3.1058
12	0.8726	1.3562	1.7823	2.1788	2.6810	3.0545
13	0.8702	1.3502	1.7709	2.1604	2.6503	3.0123
14	0.8681	1.3450	1.7613	2.1448	2.6245	2.9768
15	0.8662	1.3406	1.7531	2.1314	2.6025	2.9467
16	0.8647	1.3368	1.7459	2.1199	2.5835	2.9208
17	0.8633	1.3334	1.7396	2.1098	2.5669	2.8982
18	0.8620	1.3304	1.7341	2.1009	2.5524	2.8784
19	0.8610	1.3277	1.7291	2.0930	2.5395	2.8609
20	0.8600	1.3253	1.7247	2.0860	2.5280	2.8453
21	0.8591	1.3232	1.7207	2.0796	2.5176	2.8314
22	0.8583	1.3212	1.7171	2.0739	2.5083	2.8188
23	0.8575	1.3195	1.7139	2.0687	2.4999	2.8073
24	0.8569	1.3178	1.7109	2.0639	2.4922	2.7969
25	0.8562	1.3163	1.7081	2.0595	2.4851	2.7874
26	0.8557	1.3150	1.7056	2.0555	2.4786	2.7787
27	0.8551	1.3137	1.7033	2.0518	2.4727	2.7707
28	0.8546	1.3125	1.7011	2.0484	2.4671	2.7633
29	0.8542	1.3114	1.6991	2.0452	2.4620	2.7564
30	0.8538	1.3104	1.6973	2.0423	2.4573	2.7500
40	0.8507	1.3031	1.6839	2.0211	2.4233	2.7045
60	0.8477	1.2958	1.6706	2.0003	2.3901	2.6603
120	0.8446	1.2886	1.6577	1.9799	2.3578	2.6174
∞	0.8416	1.2816	1.6449	1.9600	2.3263	2.5758

Example: $t_{0.975;\,15} = 2.1314$ (two-sided test, $\alpha = 0.05$, $n = 16$). Symmetry: $t_{1-p;\,\mathrm{df}} = -t_{p;\,\mathrm{df}}$. The row ∞ is the standard normal distribution.

Table 2: Quantiles $t_{p;\,\mathrm{df}}$ of the t-distribution with df degrees of freedom